



10.06.2026

Response to QSK50 Engine Issue

Dear Colleagues,

Thank you for choosing NHL products and for your trust in our brand over many years. We would like to inform you of the conclusions reached by our company's specialists following a series of studies regarding the failure of QSK50 internal combustion engines (ICEs) at the Magadan project:

1. The class of NTE200 engines used meets Tier2 emission standards with an MCRS fuel system and is very sensitive to fuel and oil quality. Tests conducted by our engineers at the project site and in our laboratories at the plant in China have shown negative results – the composition of deposits indicates a significant amount of elements pointing to high oil ash content, caused by an excessive quantity of additives in the oil. This oil does not meet Cummins requirements for this engine type. Deposits and oxidation products found during injector disassembly indicate excessive contamination and water in the fuel, and require regular fuel analysis to ensure compliance with the engine manufacturer's recommendations.
2. Regarding claims of additional engine boosting, we hereby inform you that the power consumption of the GE drive system has not been altered during any software modifications. The ICE software also does not provide for additional boosting.
3. On QSK50 engines, the exhaust valves are manufactured from high-strength chromium-nickel steels without additional STELLITE hard-facing on the part of the valve face that contacts the valve seat. Under rated operating conditions, this ensures high reliability and time between overhauls.

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4. High dust levels in the quarry necessitate frequent servicing of air filters, which can lead to their damage, reduced performance, and ingress of contaminants into the ICE system. We recommend replacing filters with new ones every 500 engine hours. We do not recommend blowing them out more than once.

5. We recommend using only manufacturer-approved coolant in the engine cooling system. Furthermore, mixing coolants of different classes is strictly prohibited!

6. We recommend introducing the following:

- Mandatory inspection and adjustment of engine valves upon reaching 1500 operating hours for new ICEs, and clearance checks every 5000 operating hours.
- Endoscopic inspection of cylinders when deviations are found during valve adjustment, or when other ICE related faults are identified.
- Cylinder head service every 12,500 operating hours, or as necessary.

7. We strongly recommend against using replacement ICEs supplied by third parties without NHL's approval, as this may lead to failures of associated components (particularly the generator) and improper operation of the dump truck due to possible incorrect ICE calibration, as well as potential cases of marine or energy ICEs being converted for mining conditions using parts not suitable for such loads. We inform you that if any use of the above-mentioned ICEs is detected, NHL will not support the warranty on the GE drive system.

NHL informs you that to monitor and improve the current situation regarding QSK50 engine failures, we have taken the following measures:

- Strengthening the Magadan project team with additional ICE specialists;
- Sending an additional stock of necessary spare parts to minimize downtime in case of failures;
- Providing daily consultations and monitoring the situation together with the project and the engine manufacturer.



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We are confident that if the above recommendations are followed, the number of engine failures will decrease significantly. To discuss joint future steps to improve the current availability factor of the NTE200 machines at the Magadan project, we propose holding a joint trilateral meeting during the week of June 8–12.

Best Regards !

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